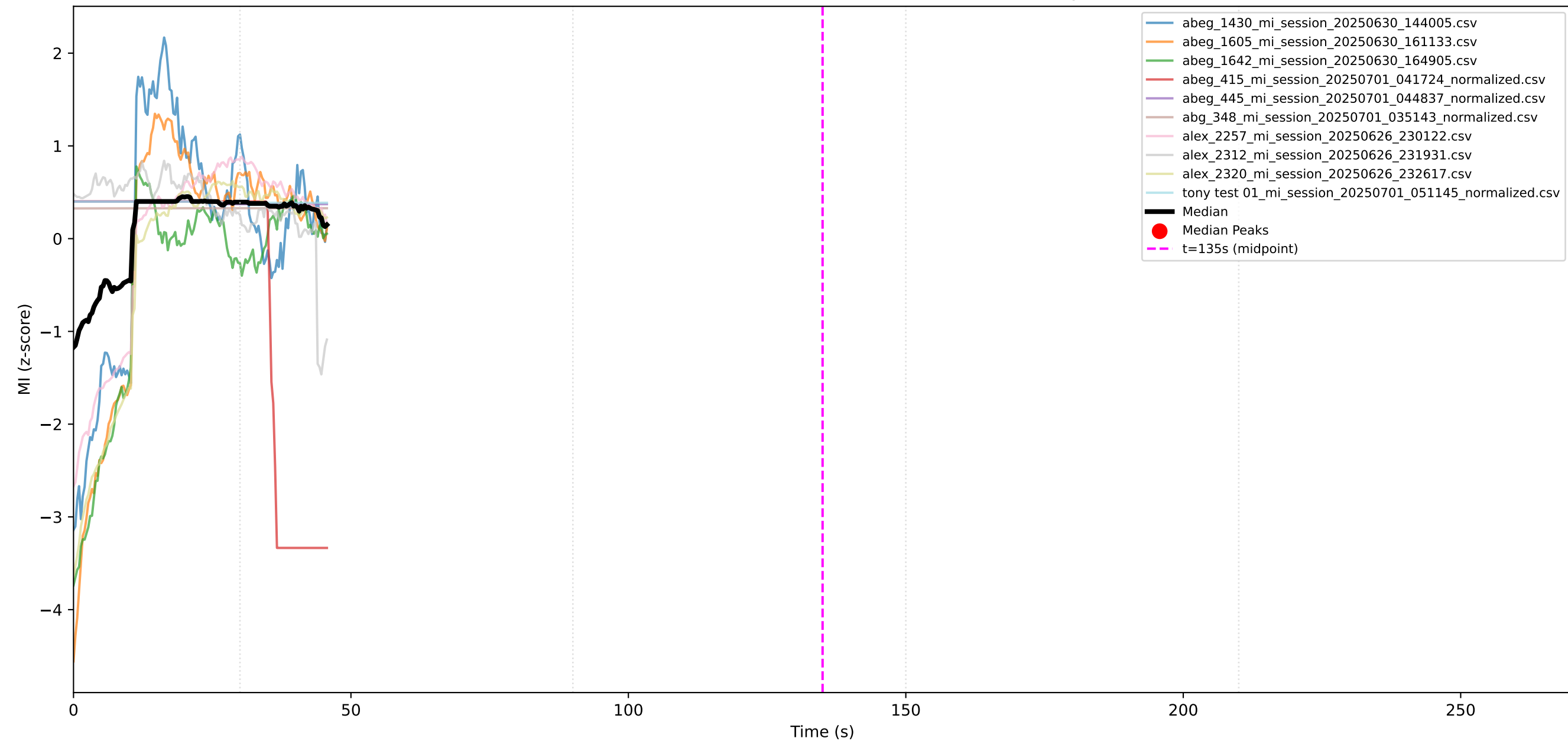
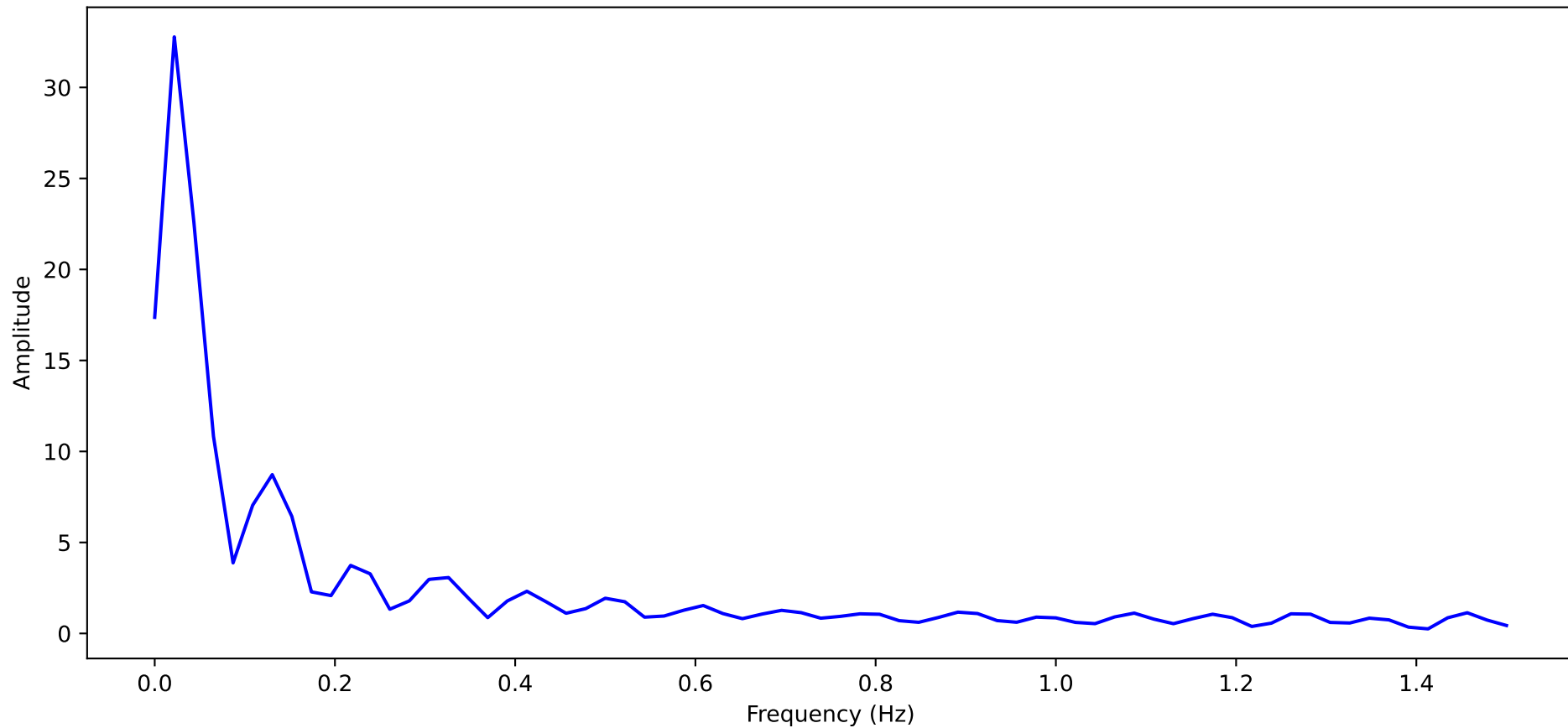


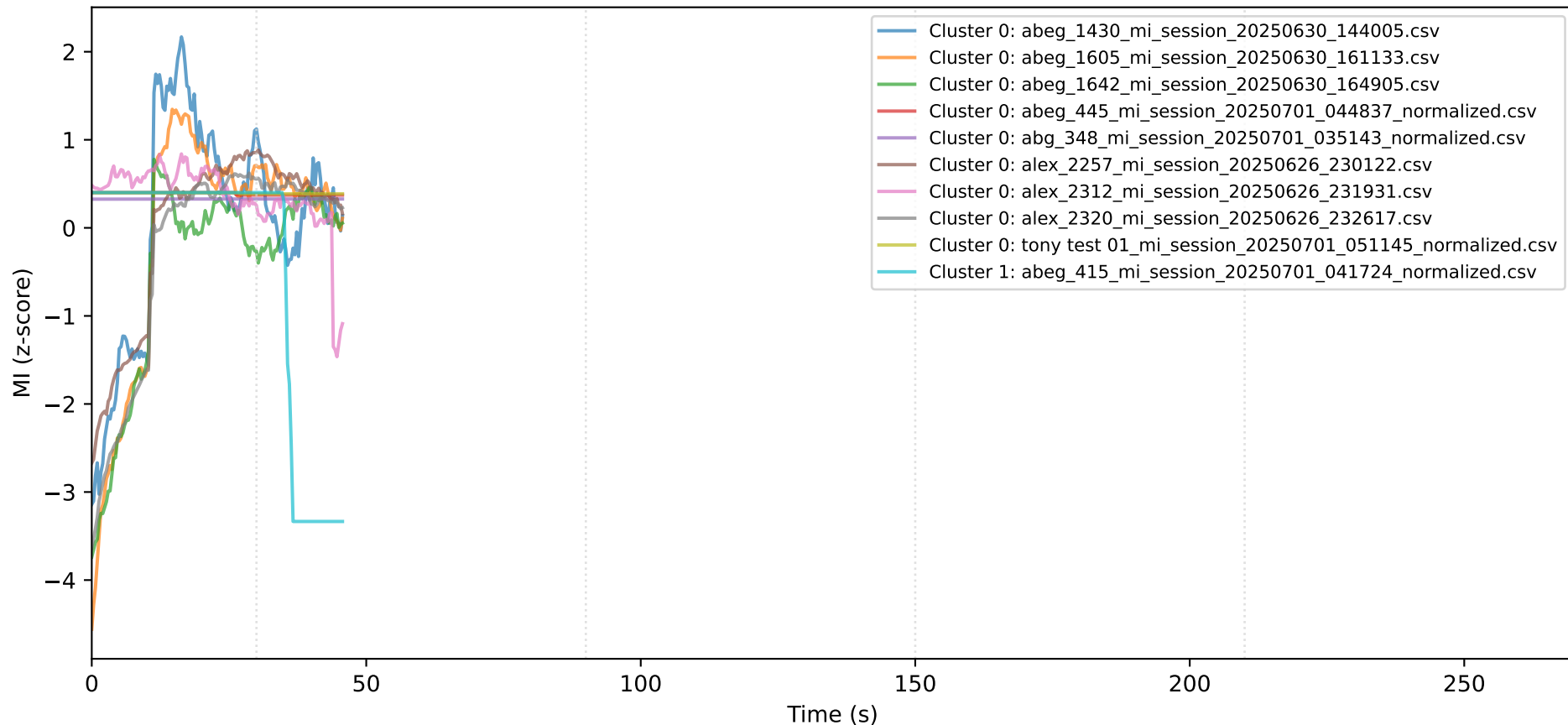
MI Curves (z-score, MA=20) with Median and Peaks (270s @ 3Hz)



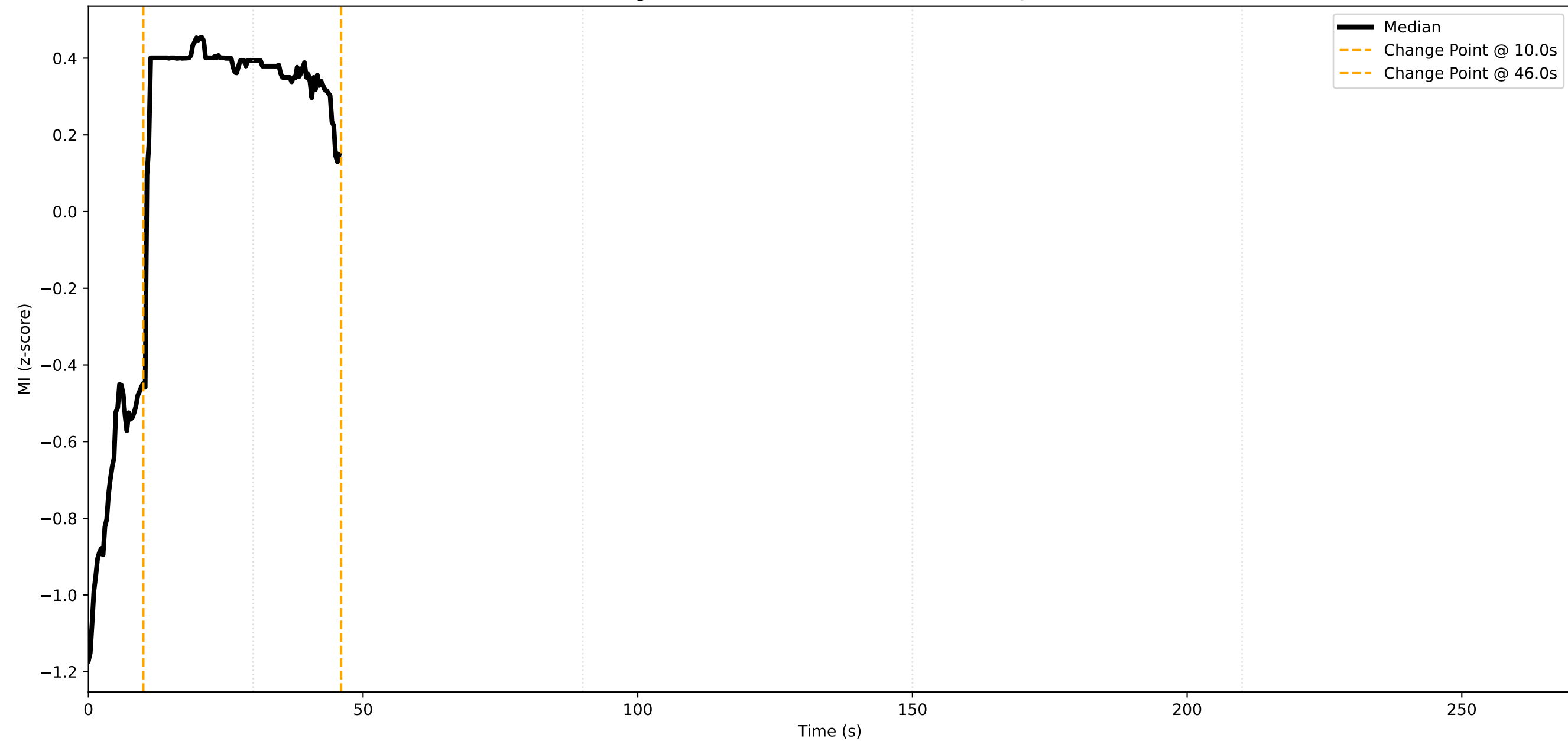
Median MI Frequency Spectrum (FFT)



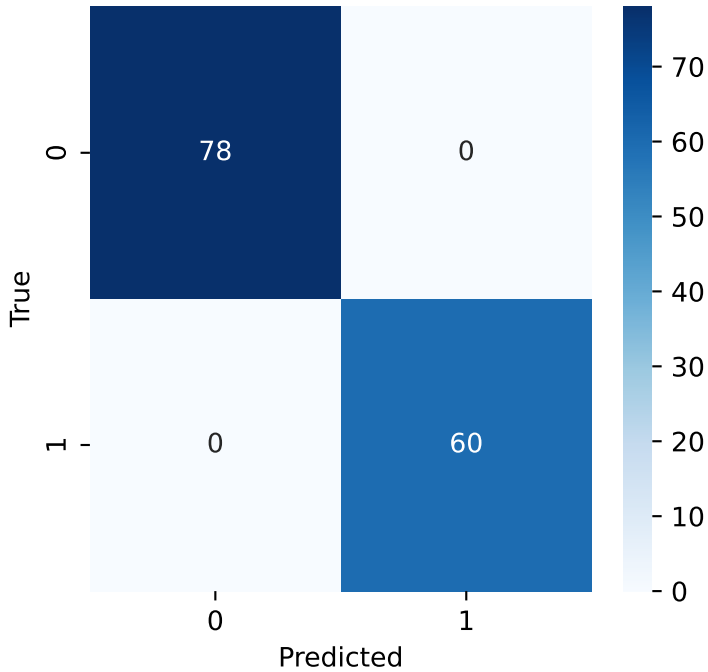
KMeans Clustering of MI Curves (k=2) - 270s @ 3Hz



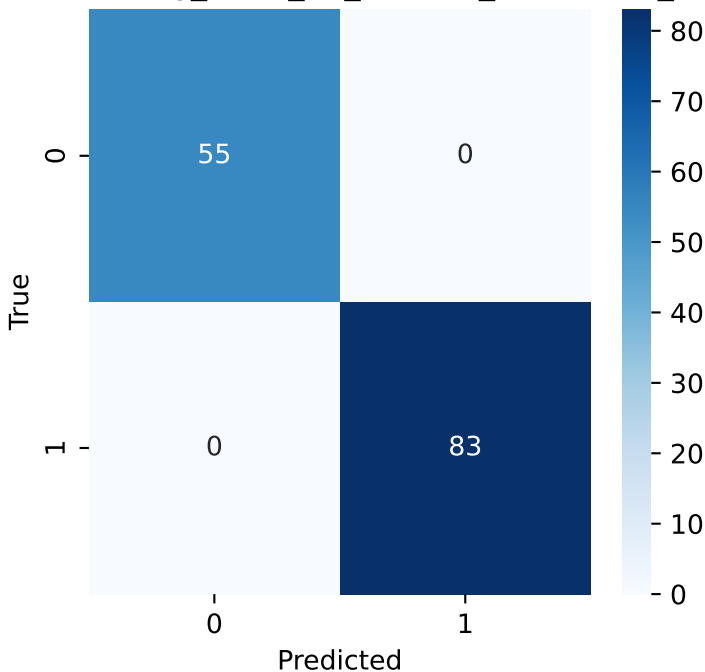
Change Point Detection on Median MI - 270s @ 3Hz



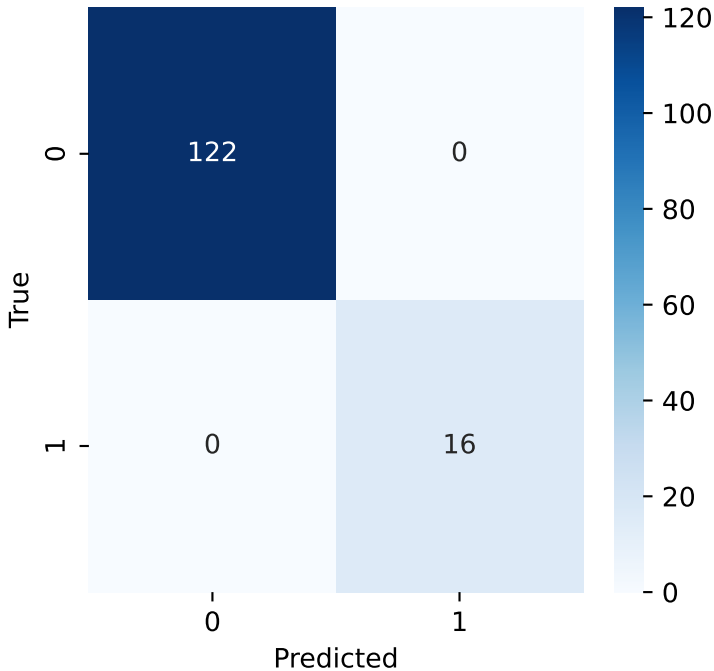
Matrix: abeg_1430_mi_session_20250630_1440



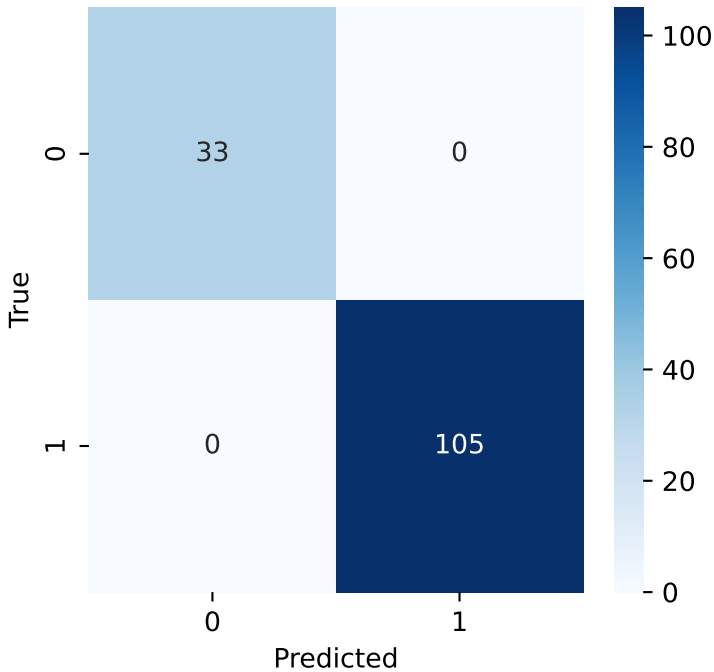
Matrix: abeg_1605_mi_session_20250630_1611



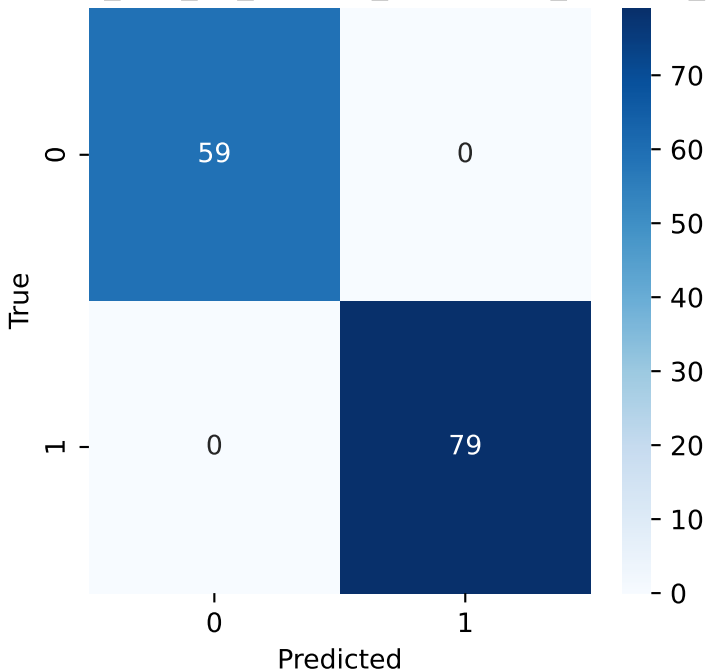
Matrix: abeg_1642_mi_session_20250630_1649



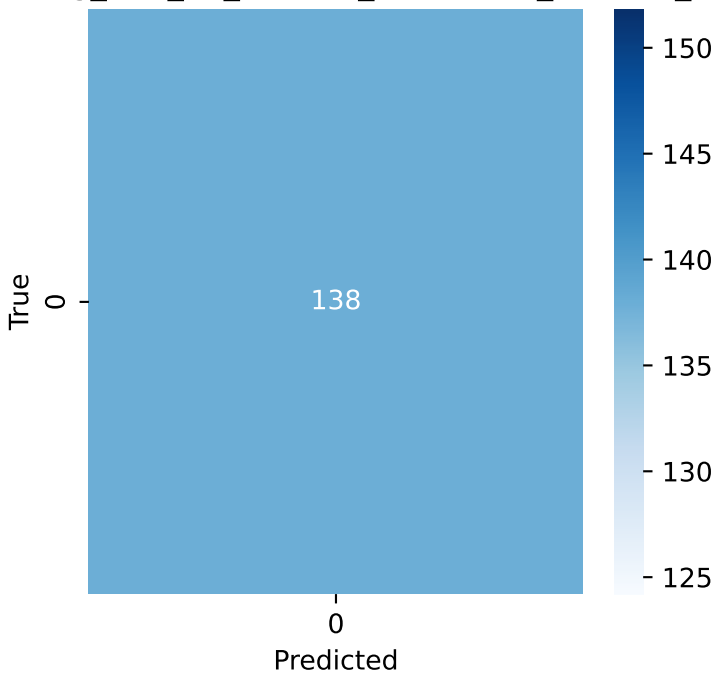
x: abeg_415_mi_session_20250701_041724_no



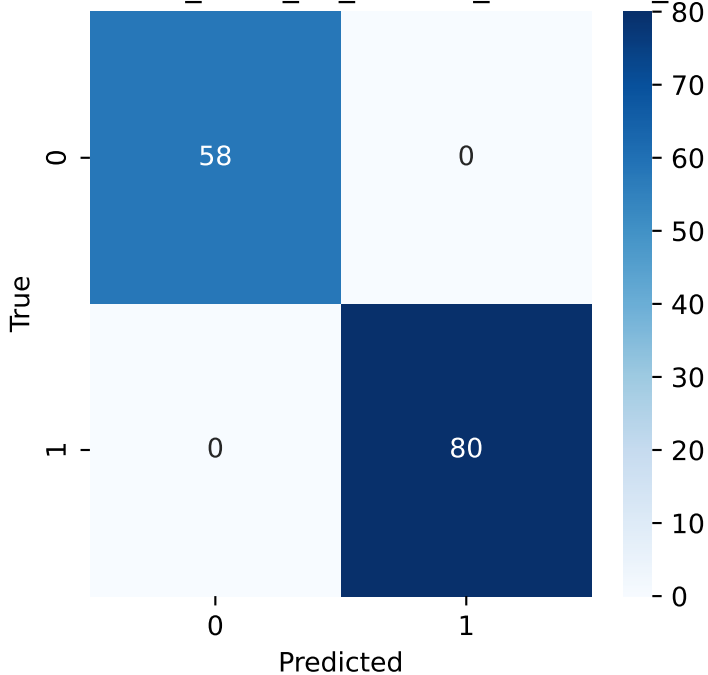
ix: abeg_445_mi_session_20250701_044837_no



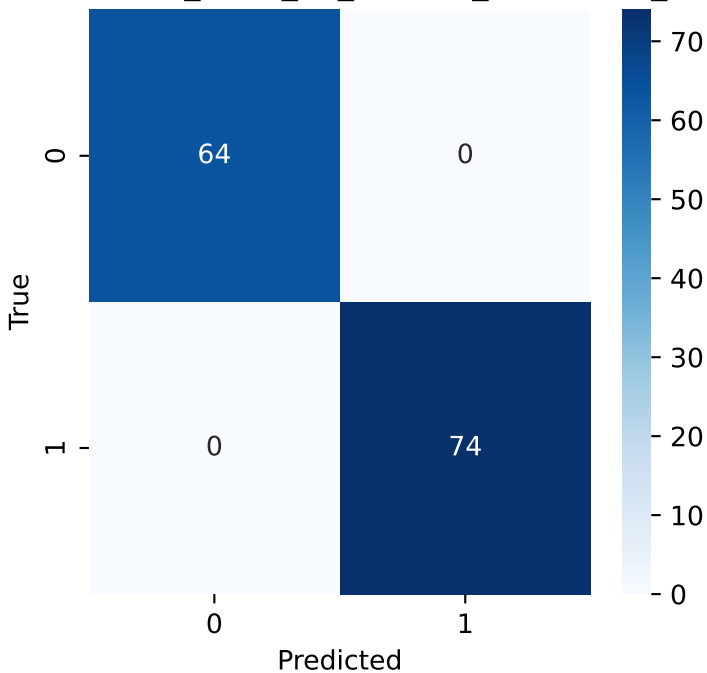
ix: abg_348_mi_session_20250701_035143_nor



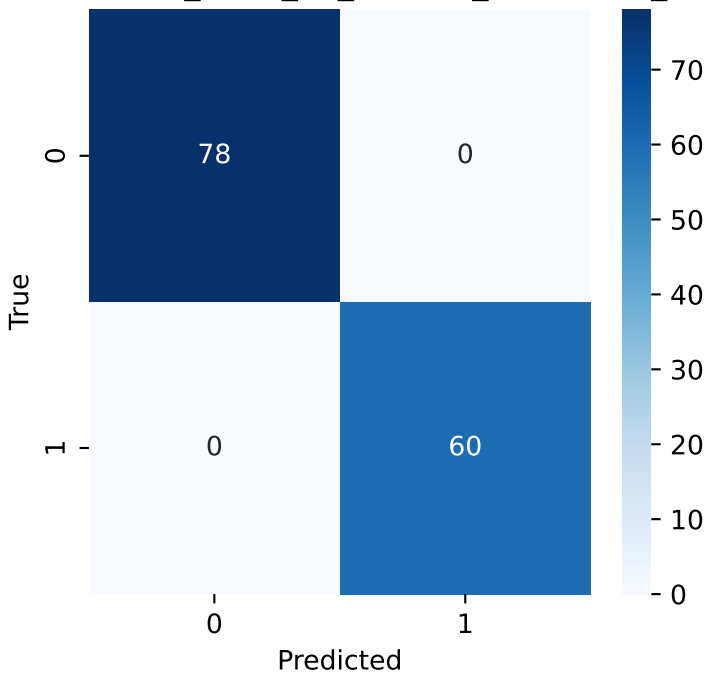
h Matrix: alex_2257_mi_session_20250626_2301



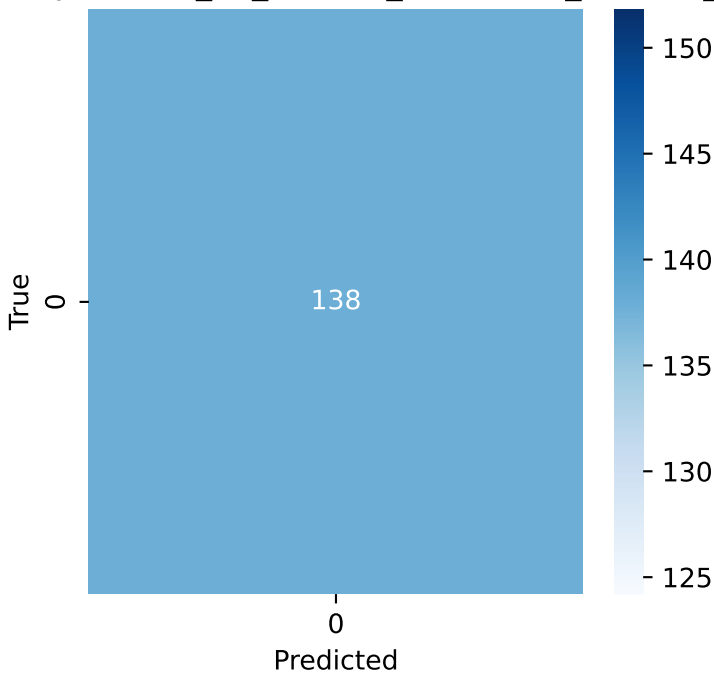
h Matrix: alex_2312_mi_session_20250626_2319



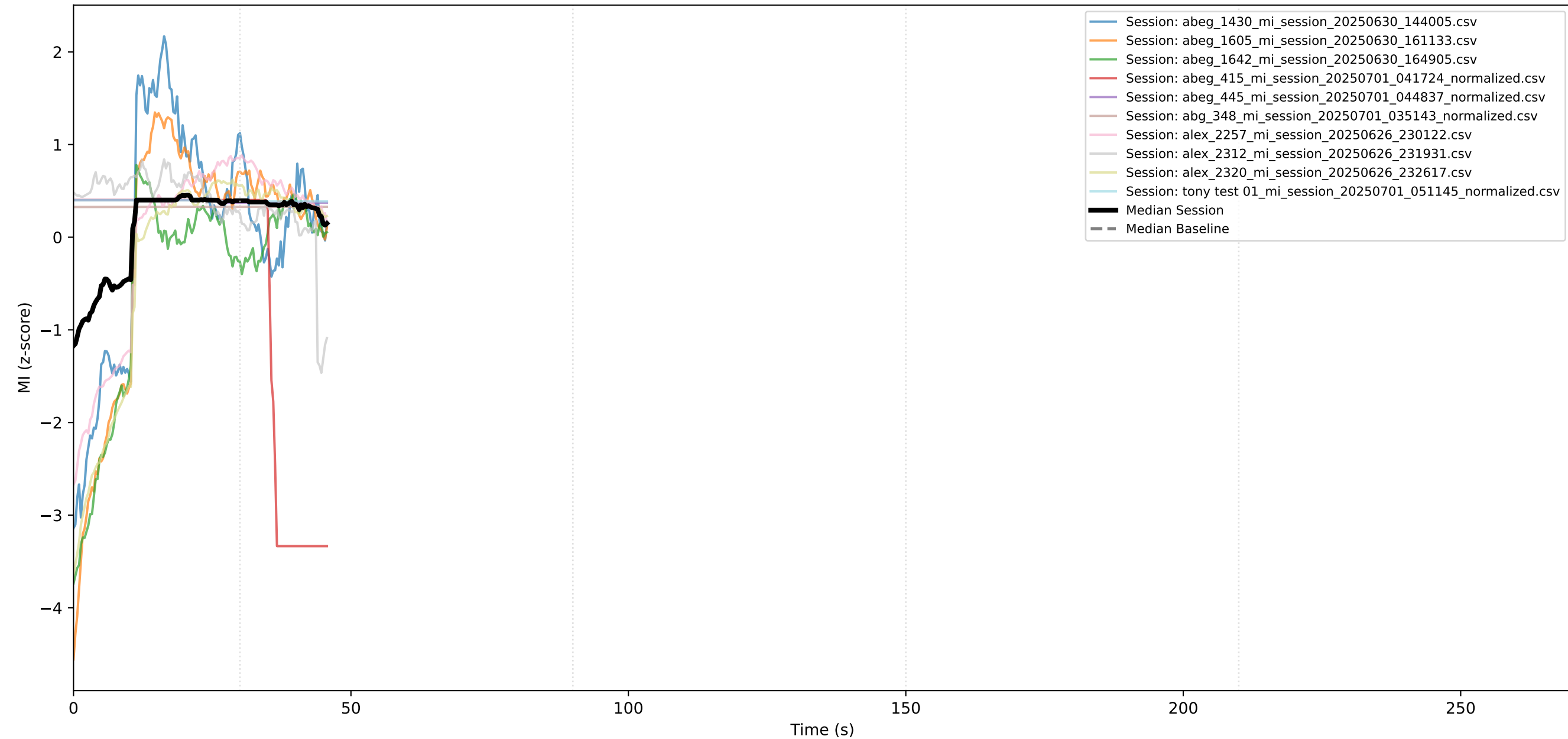
h Matrix: alex_2320_mi_session_20250626_2326



: tony test 01_mi_session_20250701_051145_no



Session MI (up to 10) vs. Median Baseline (Eyes Closed) - 270s @ 3Hz



MI Advanced Report

Generated: 20250717_025257

MI Advanced Report - Statistical Summary (3Hz Sampling)

1. Mindfulness Index (MI) Trajectory

- * All MI session files with ≥ 300 samples were included (100s minimum at 3Hz).
- * Individual and median MI curves were plotted with 20-second moving average (see Figure 8.1).
- * RM-ANOVA across six epochs (0-45s, 45-90s, 90-135s, 135-180s, 180-225s, 225-270s):
Anova

	F Value	Num DF	Den DF	Pr > F
Epoch	nan	5.0000	45.0000	nan

- * Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
0-45s	135-180s	nan	nan	nan	nan	False
0-45s	180-225s	nan	nan	nan	nan	False
0-45s	225-270s	nan	nan	nan	nan	False
0-45s	45-90s	-0.3182	nan	nan	nan	False
0-45s	90-135s	nan	nan	nan	nan	False
135-180s	180-225s	nan	nan	nan	nan	False
135-180s	225-270s	nan	nan	nan	nan	False
135-180s	45-90s	nan	nan	nan	nan	False
135-180s	90-135s	nan	nan	nan	nan	False
180-225s	225-270s	nan	nan	nan	nan	False
180-225s	45-90s	nan	nan	nan	nan	False
180-225s	90-135s	nan	nan	nan	nan	False
225-270s	45-90s	nan	nan	nan	nan	False
225-270s	90-135s	nan	nan	nan	nan	False
45-90s	90-135s	nan	nan	nan	nan	False

- * Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- * Peak analysis (median curve): 0 peaks ($z > 1.5$) at sample times [], amplitudes [], durations [] (see Table 8.2, Figure 8.4).
- * AUC (median): 17.88; mean $\pm$ SD (individual): 4.29 $\pm$ 43.48 (see Table 8.3).
- * Bootstrap 95% CI for mean MI difference (0-135s vs 136-270s): mean diff = 0.000, CI = [0.000, 0.000].
- * Change point detection (Pelt algorithm): 1 change points at sample indices [30] (see Figure 8.5).

3. Clustering & Mixed-Effects

- * K-Means clustering (k=2): cluster averages {}, user distribution {0: 9, 1: 1} (see Figure 8.6).
- * Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

Model:	MixedLM	Dependent Variable:	value			
No. Observations:	1380	Method:	REML			
No. Groups:	10	Scale:	0.9292			
Min. group size:	138	Log-Likelihood:	-1928.3396			
Max. group size:	138	Converged:	Yes			
Mean group size:	138.0					

	Coef.	Std.Err.	z	P> z	[0.025	0.975]

Intercept	-0.408	0.116	-3.519	0.000	-0.635	-0.181
time	0.006	0.001	9.692	0.000	0.005	0.008
file Var	0.108	0.056				